

Contest Discussion

01. Chef & Cooking
02. Benjamin & XOR
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Chef and Cooking

Problem Description

In a parallel universe, there exist a kingdom that is known for its unique way of cooking. In this kingdom, there is a famous chef who is known for her delicious dishes. One day, the chef decided to create a new dish that consists of a sequence of ingredients. Each ingredient has a distinct weight, and the chef wants to choose a subarray of ingredients that have an increasing weight.

The chef wants to know the maximum possible sum of the weights of the ascending subarray she can choose. Can you help the chef by writing a function that returns the maximum possible sum of an ascending subarray in the weights of ingredients?

The array of ingredients is represented by the array A .

$$A = \{1, 2, 3, 4, 5\} \quad \text{Ans} = 15$$

$$A = \{2, 4, 3, 6, 7\} \quad \text{Ans} = 16$$

6 16

$$A = \{9, 2, 4, 2\} \quad \underline{\underline{\text{Ans} = 9}}$$

9 6 2

$$A = \{2, 2, 3\} \quad \underline{\underline{\text{Ans} = 5}}$$

2 3

```
long solve(int [] A)
```

```
    long ans = A[0] ,
```

```
    long currsum = A[0]
```

```
    for (i = 1; i < n; i++) {
```

```
        if (A[i] > A[i-1]) {
```

```
            currsum += A[i];
```

```
        } else {
```

```
            currsum = A[i];
```

```
        ans = Math.max(ans, currsum);
```

```
    }
```

```
}
```

A = { 1, 1, 6, 11, 14, 14, 17, 18, 18, 1 }

1 32 49 18 1

Ans = 49

* Benjamin & XOR

Problem Description

In the picturesque town of Bitville, nestled amidst rolling hills and shimmering lakes, the residents had a deep fascination with the world of bits and binary operations. The townsfolk were avid mathematicians, always seeking new puzzles to unravel and insights to gain.

One sunny morning, the town's esteemed professor of mathematics, Dr. Benjamin, presented a captivating challenge to his students. He introduced them to an array **A** of **N** integers, representing a sequence of numbers with each element holding a special significance.

Dr. Benjamin explained that the students' task was to analyze the array and determine the count of pairs that satisfied a unique condition. The condition revolved around the XOR operation on the **i**th bit of the pair's elements. The goal was to **count the pairs** for which the **xor** of the **i**th bit resulted in **one**. You have to answer for **Q** queries given in array **B**, each query denotes the index for which you need to find the count of pairs with xor of that index equals 1.

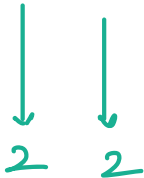
Can you solve the task given by Dr. Benjamin?

Please read the examples given below for better understanding of the problem.

count of pairs $\rightarrow$ xor of i^{th} bit resulted in one

$$A = \{1, 2, 3\}$$

$$B = \{0, 1\}$$



$$1 = 001$$

$$2 = 010$$

$$3 = 011$$

$$\left. \begin{array}{l} 1 \wedge 2 \\ 2 \wedge 3 \end{array} \right\} 0^{th} \text{ bit set}$$

$$\left. \begin{array}{l} 1 \wedge 2 \\ 1 \wedge 3 \end{array} \right\} 1^{st} \text{ bit set}$$

$$A = \{2, 4, 7, 11\}$$

$$B = \{3, 1\}$$



Count the set bit at 3^{rd} idx
Count the unset bit at 3^{rd} idx

2	$\rightarrow$	0	0	1	0
4	$\rightarrow$	0	1	0	0
7	$\rightarrow$	0	1	1	1
11	$\rightarrow$	1	0	1	1

$$2 \wedge 11$$

$$4 \wedge 11$$

$$7 \wedge 11$$

$$A = \{ \text{ele} \}$$

$$B = \text{queries} =$$

```
int [] ans = new int [B.length];
```

```
for (j = 0; j < B.length; j++)
```

```
    int k = B[j];
```

```
    int cset = 0
```

```
    int cunset = 0
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (ar[i] & (1 << k) != 0) {
```

```
            cset ++;
```

```
        } else {
```

```
            cunset ++;
```

```
        }  
    }  
    ans[j] = cset * cunset;
```

```
}
```

* Rain water Trapped

lmax = 0 rmax = 0

i = 0 , j = n-1

while (i < j) {

lmax = Math.max(lmax, ar[i])

rmax = Math.max(rmax, ar[j])

if (lmax < rmax) {

ans += lmax - ar[i];

i++;

2

else {

ans += rmax - ar[j];

j--;

3

3

Tc: O(n)
Sc: O(1)

Effective time { $\rightarrow$ use less submission
 $\rightarrow$ less time